

Varshitha Ramakumar

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PROFILE

M.Sc. student in Aerospace Engineering at TU Munich with a B.Sc. in Mechanical Engineering (specialisation: Energy Engineering) from KIT. Hands-on experience in materials characterisation through a two-year research role at IAM-WK (KIT) and an independently conducted bachelor's thesis: XRD profile analysis, SEM/EDX microstructure analysis, DSC, Vickers hardness testing, and mechanical compression testing on Ni- and Ti-base alloys. Familiar with structured laboratory work, data analysis, technical documentation, and working in an academic research environment. Strong self-initiative and experience in independent experimental planning.

EDUCATION

M.Sc. Aerospace Engineering

Munich, Germany Apr. 2026 – present

Technical University of Munich (TUM)

- Relevant coursework: Aircraft Design, Spacecraft Design, Gas Dynamics, Applied CFD, Aircraft Systems.

B.Sc. Mechanical Engineering (International) – Specialisation: Energy Engineering

Karlsruhe, Germany Oct. 2022 – Mar. 2026

Karlsruhe Institute of Technology (KIT)

- Core subjects: Materials Science, Manufacturing Technology, Thermodynamics, Engineering Mechanics, Combustion, Advanced Mathematics, Computer Science.
- Design project: creation of complete engineering drawings to DIN/ISO standards.
- **Bachelor's thesis:** “*The Effect of Al-Addition on Phase Stability of Ti-Rich RCCAs*” – Supervisor: Amin Radi; Examiners: Prof. Dr.-Ing. Martin Heilmaier, Dr.-Ing. Daniel Schliephake (IAM-WK, KIT, Dec. 2025). Investigation of three Ti-rich Ta–Mo–Ti–Cr–Al alloys (6–10 at.% Al) after ageing at 800 °C and 1000 °C (10–1000 h) using SEM/EDX, XRD (Nelson–Riley, $a \approx 3.170 \text{ \AA}$), DSC (Netzsch DSC 404 F1 Pegasus), Vickers hardness testing (HV1), and compression testing (ZwickRoell Z100, $\approx 1150\text{--}1300 \text{ MPa}$ at RT, $750\text{--}850 \text{ MPa}$ at 800 °C). Key finding: stable single-phase A2 bcc matrix across all conditions; no evidence of an A2 → B2 transformation; CALPHAD predictions (Pandat/PanRHEA2025) correlated with experimental results.

Grade 12 – All India Senior School Certificate Examination (AISSCE)

Chennai, India 2021

Vidya Mandir Sr. Secondary School

- Final grade: 95.8 % Subjects: Mathematics, Physics, Chemistry, Computer Science (Python), English.

WORK EXPERIENCE

Student Research Assistant (HiWi) – Materials Characterisation & FEM Simulation

Karlsruhe, Germany May 2023 – Sep. 2025

Institute for Applied Materials – Materials Science (IAM-WK), KIT

- **XRD profile analysis:** peak fitting on approx. 200 samples of nickel and aluminium alloys using P61A – a Python-based evaluation software developed at KIT in collaboration with a research laboratory in Aachen; method selection, source-code bug-fixing, systematic evaluation, and structured documentation of results.
- **FEM simulation in ABAQUS** on Inconel 718 (Ni-base superalloy): stress–strain analysis under mechanical and thermal loading; modelling of the wire electrical discharge machining (WEDM) process to analyse machining behaviour.
- Independent planning and execution of a two-year work programme: test plans, experimental procedures, data acquisition, and structured documentation; regular coordination with project leads and subject-matter experts.
- Data analysis and processing with Python; preparation of reports and presentations for results communication.

Tutor – Materials Science I/II

Karlsruhe, Germany Sep. 2025

Karlsruhe Institute of Technology (KIT)

- Conducted tutorials on selected topics from the Materials Science I/II curriculum; administered colloquia to assess students' subject understanding.

- Supervised the laboratory experiment on time–temperature transformation of Pb–Sn alloys; guided students through the procedure and discussed the significance of precipitation hardening.

Supply Chain Planning & Engineering Intern

Sinsheim, GermanyNov. 2025 – Apr. 2026

Gebhardt Fördertechnik GmbH

- Developed facility layout designs in Autodesk Inventor and Revit; managed engineering documents and CAD data using Autodesk Vault (PDM).
- Built and simulated digital twins of manufacturing and intralogistics systems in Emulate3D for virtual commissioning.
- Close collaboration with manufacturing, production planning, and project management; structured project documentation.

PROJECTS & RESEARCH

Structural Properties of High-Temperature Superconductors (YBCO)

Karlsruhe, GermanyJul. 2024

IAM-WK, KIT

- Literature review and academic presentation on the structure and physical properties of YBCO as part of the course “Scientific Methods in Mechanical Engineering” (Supervisor: Dr. Mazzocchi, IAM-WK).

CERTIFICATIONS

- **Advanced CAD/PDM Certificate** – Institute of Product Engineering, KIT (PTC Creo Parametric 8.0, PDM Windchill) Dec. 2023
- **Basic CAD Certificate** – Institute of Product Engineering, KIT (PTC Creo Parametric 8.0) Jul. 2023
- **MATLAB Onramp** – MathWorks (data import, numerical analysis, visualisation) Dec. 2023

SKILLS

Materials Char.	SEM/EDX (BSE microstructure analysis, phase identification), XRD (profile analysis, Nelson–Riley method, P61A incl. bug-fixing), DSC, Vickers hardness testing (HV1), compression testing (ZwickRoell)
Alloy Systems	Ni-base superalloys (Inconel 718), Ti-rich refractory alloys (Ta–Mo–Ti–Cr–Al RCCAs)
Therm. Analysis	Interpretation of CALPHAD phase diagrams (Pandat/PanRHEA2025); correlation of thermodynamic predictions with experimental results
Simulation & FEA	ABAQUS (stress–strain analysis, thermomechanical loading, WEDM modelling)
Data Analysis	Python (XRD evaluation pipeline, data processing, bug-fixing), MATLAB (numerical analysis, visualisation), SQL
CAD & PDM	PTC Creo Parametric 8.0, Autodesk Inventor, Autodesk Fusion 360, AutoCAD; Autodesk Vault, PDM Windchill
Standards	DIN, ISO (engineering drawings, tolerancing)
MS Office	Word, Excel, PowerPoint
Languages	German (B1), English (fluent), Hindi (fluent), Tamil, Telugu, Kannada (fluent)
Soft Skills	Self-initiative, structured working, teamwork, communication skills, technical writing